



FAST National University
of Computer and Emerging Sciences
Operating Systems (CS2006)

Name: Solution Quiz-2*: (CLO - 3)

Max. Marks: 02

Student ID: _____

Instructions:

Max Time: 20 Minutes

- Don't write with Pencil, Even Diagrams should be made with Pen otherwise marks will be deducted
- Instructions in Questions should be followed strictly
- Write your answer after each question -

Question 1: Solve the Following

- a) A processor has a TLB with a 85% hit rate. A TLB lookup takes 4 ns, which overlaps with the memory access cycle on a hit. Accessing main memory requires 100 ns per access.
- i) Assuming no page faults, compute the Effective Access Time (EAT) and Determine the speedup gained from using the TLB over a system with no TLB

NO
with TLB: each access = 2 memory access (page fault & data)

$$(EAT = 2 \times 100 = 200 \text{ ns})$$

with TLB: Hit = 1 memory Access = 100 ns, if Miss = 2 memory Access + TLB = 204

$$EAT = (0.85 \times 100) + (0.15 \times 204)$$

$$= 115.6 \text{ ns}$$

$$\text{Speedup} = 200 / 115.6 = 1.73$$

- b) Consider a system with a 32-bit Virtual address space, a Frame size of 4 KB, and a physical memory of 64 MB.

- i) Calculate the number of entries in the process P1 page table, assuming the entire logical address space is allocated for P1?

$$\begin{aligned} \text{V. Address space} &= 32 \text{ bit or } 2^{32} \text{ & Frame size} = 4 \text{ KB or } 2^{12} \\ &= 2^{32} / 2^{12} = 2^{20} = 1048576 \text{ entries} \end{aligned}$$

- ii) If each page table entry requires 4 bytes (including the frame number) what is the total size of a process's page table?

$$\text{Total page size} = 2^{20} \times 2^2 = 2^{22} \text{ bytes} = 4 \text{ MB.}$$

- iii) A process requires 14,500 bytes of memory. How much internal fragmentation occurs due to paging? Express your answer in bytes.

$$\text{Pages required} = 14500 / 4096 \Rightarrow 3.54 = 4 \text{ pages}$$

$$\text{Internal Fragmentation} = 4 \text{ Pages} (4 \times 4096) \Rightarrow 16,384 \text{ bytes}$$

$$= 16384 - 14500 = 1884 \text{ bytes}$$



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Question 1: Solve the Following

a) Consider a system with a 32-bit logical address space, a page size of 4 KB, and a physical memory of 64 MB.

- i) How many entries are there in a process's page table, assuming the entire logical address space is allocated?
- Virtual Address Space = 2^{32} bytes
 Page Size = 4 KB = 2^{12} bytes
 Number of entries = $\frac{2^{32}}{2^{12}} = 2^{20} = 1,048,576$ entries.

- ii) If each page table entry requires 4 bytes (including the frame number) what is the total size of a process's page table?

Page table size = $2^{20} \times 4 = 2^{22}$ bytes
 $= 4$ MB.

- iii) A process requires 15,000 bytes of memory. How much internal fragmentation occurs due to paging? Express your answer in bytes and as a percentage of the total memory allocated to the process.

Pages required = $\frac{15000}{4096} \Rightarrow 3.66 = 4$ pages
 Internal Fragmentation = $16384 - 15000 = 1,384$ bytes

b) A system has a TLB hit ratio of 95%. The time to search the TLB is 5 ns, which is included in the main memory access time in case of a hit. A main memory access takes 100 ns.

- i) Calculate the Effective Access Time (EAT) assuming there is no page fault and what is the speedup achieved by using the TLB compared to a system without a TLB (where every access requires two memory accesses)?

with NO TLB: = Each access = 2 Memory Access (Page table + data)
 $EAT = 2 \times 100 = 200$ ns.

with TLB: Hit $\rightarrow$ 1 memory access = 100 ns
 Miss: 2 memory access + TLB lookup = $4 \times 200 = 204$ ns

$EAT = (0.95 \times 100) + (0.05 \times 204)$
 $= 95 + 10.2 = 105.2$ ns